

USER MANUAL

HWT9053(RS485)

IP67 Inclinometer





Tutorial Link

[Google Drive](#)

Link to instructions DEMO:

[WITMOTION Youtube Channel](#)

[HWT9053 RS485 Playlist](#)

If you have technical problems or cannot find the information that you need in the provided documents, please contact our support team. Our engineering team is committed to providing the required support necessary to ensure that you are successful with the operation of our AHRS sensors.

Contact

[Technical Support Contact Info](#)

Application

- AGV Truck
- Platform Stability
- Auto Safety System
- 3D Virtual Reality
- Industrial Control
- Robot
- Car Navigation
- UAV
- Truck-mounted Satellite Antenna Equipment

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1 Introduction

The HWT9053 integrates high-precision gyroscopes, accelerometers, and MMC3630 magnetic field sensors. It uses high-performance microprocessors and advanced dynamics calculation and Kalman dynamic filtering algorithms to quickly solve the current real-time motion posture of the module.

HWT9053 offers several advantages over competing sensor:

- The use of advanced digital filtering technology can effectively reduce measurement noise and improve measurement accuracy.
- The module integrates an attitude solver, with the dynamic Kalman filter algorithm, can accurately output the current attitude of the module in a dynamic environment, the attitude measurement accuracy is 0.001° (HWT9052 is 0.01°), the stability is extremely high, and the performance is even better than a certain Some professional inclinometers
- The module has its own voltage stabilization circuit, and the working voltage is 5V~36V.
- Support 485 interface. It is convenient for users to choose the best connection method. The serial port rate is adjustable from 4800bps to 921600bps.
- 4-layer PCB board technology, thinner, smaller and more reliable.

1.1 Warning Statement

- Putting more than 36 Volt across the sensor wiring of the main power supply can lead to permanent damage to the sensor.
- VCC cannot connect with GND directly, otherwise it will lead to the burning of the circuit board.
- For proper instrument grounding: use WITMOTION with its original factory-made cable or accessories.
- Do not access the I2C interface.
- For secondary developing project or integration: use WITMOTION with its compiled sample code.

2 Use Instructions with PC

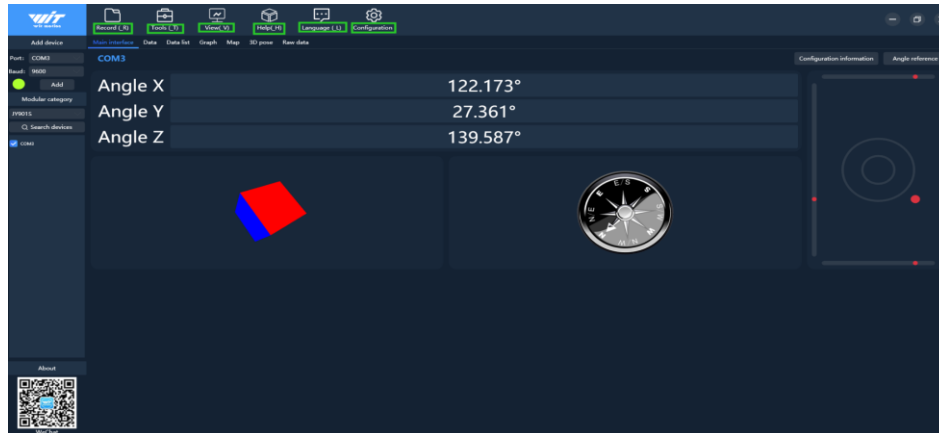
Hit the hyperlink direct to the document or download center:

- [Software and driver download](#)
- [Quick-guide Manual](#)
- [Teaching Video](#)
- [Common Software with detailed instructions](#)
- [SDK\(sample code\)](#)
- [SDK Tutorial Documentation](#)
- [Communication Protocol](#)

3 Software Introduction

[Software function introduction](#)

(Ps. You can check the functions of the software menu from the link.)



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4 MCU Connection

Step 1. Connect the sensor with a serial converter

PIN Connection:

VCC - 5~36V

B - B

A - A

GND - GND

(VCC 5~36V is recommended for connection)

RS485 PIN DEFINITION

RED	YELLOW	GREEN	BLACK
VCC 5V-36V	A	B	GND

